

Given (for all proofs below)

$V = IR + E_{mf}$ by Ohm's Law with back EMF for a DC motor

$E_{mf} = \omega \kappa_e$ by the Back EMF equation of a motor

$\tau = I \kappa_t$ by the Motor torque equation

$\tau = J\alpha + B\omega + \tau_{load} \text{sign}(\omega)$ by the Rotational equation of motion

The value of ω and α (Only for the proof directly below this)

Find a function V_{ff} , voltage, for a feedforward controller

$V_{ff} := V$ for this proof only for notational ease of use

$\omega := \omega_{ref}$ for this proof only for notational ease of use

$\alpha := \alpha_{ref}$ for this proof only for notational ease of use

$V = IR + E_{mf}$, $E_{mf} = \omega \kappa_e$, $\tau = I \kappa_t$, $\tau = J\alpha + B\omega + \tau_{load} \text{sign}(\omega)$, and the value of ω and α because they are given

$I \kappa_t = J\alpha + B\omega + \tau_{load} \text{sign}(\omega)$ By substitution

$I = \frac{J\alpha + B\omega + \tau_{load} \text{sign}(\omega)}{\kappa_t}$ By Division

$-IR = E_{mf} - V$ By Subtraction

$I = \frac{E_{mf} - V}{-R}$ By Division

$\frac{E_{mf} - V}{-R} = \frac{J\alpha + B\omega + \tau_{load} \text{sign}(\omega)}{\kappa_t}$ By substitution

$\frac{\omega \kappa_e - V}{-R} = \frac{J\alpha + B\omega + \tau_{load} \text{sign}(\omega)}{\kappa_t}$ By substitution

$\omega \kappa_e - V = \frac{J\alpha + B\omega + \tau_{load} \text{sign}(\omega)}{\kappa_t} (-R)$ By multiplication

$-V = \frac{J\alpha + B\omega + \tau_{load} \text{sign}(\omega)}{\kappa_t} (-R) - \omega \kappa_e$ By subtraction

$V = -\left(\frac{J\alpha + B\omega + \tau_{load} \text{sign}(\omega)}{\kappa_t} (-R) - \omega \kappa_e \right)$ By division

$V = \frac{J\alpha + B\omega + \tau_{load} \text{sign}(\omega)}{\kappa_t} (R) + \omega \kappa_e$ By substitution

$V = \frac{J\alpha R}{\kappa_t} + \frac{B\omega R}{\kappa_t} + \frac{\tau_{load} \text{sign}(\omega) R}{\kappa_t} + \frac{\omega \kappa_e \kappa_t}{\kappa_t}$ By substitution

$V = \frac{J\alpha R}{\kappa_t} + \frac{B\omega R + \omega \kappa_e \kappa_t}{\kappa_t} + \frac{\tau_{load} \text{sign}(\omega) R}{\kappa_t}$ By substitution

$V = \frac{J\alpha R}{\kappa_t} + \frac{\omega(BR + \kappa_e \kappa_t)}{\kappa_t} + \frac{\tau_{load} \text{sign}(\omega) R}{\kappa_t}$ By substitution

Define K_v by $K_v = \frac{BR + \kappa_e \kappa_t}{\kappa_t}$

Define K_a by $K_a = \frac{JR}{\kappa_t}$

Define K_s by $K_s = \frac{\tau_{load} R}{\kappa_t}$

$V = K_a \alpha + K_v \omega + K_s \text{sign}(\omega)$ by substitution

$V_{ff} = K_a \alpha_{ref} + K_v \omega_{ref} + K_s \text{sign}(\omega_{ref})$ by substitution

Define a function V_{ff} by $V_{ff}(\omega_{ref}, \alpha_{ref}) = K_a \alpha_{ref} + K_v \omega_{ref} + K_s \text{sign}(\omega_{ref})$

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Given (Only for the proof directly below this)

The value of ω

Find the maximum α angular acceleration at some angular velocity ω

$V = IR + E_{mf}$, $E_{mf} = \omega \kappa_e$, $\tau = I \kappa_t$, $\tau = J\alpha + B\omega + \tau_{load} \text{sign}(\omega)$, and the value of ω because they are given

Note: The VEX motor controller has current limits so we must also account for those.

Define $I_{\min}$ and $I_{\max}$ as the current limits and define I_{clamp} as the clamped value of I

Define $V_{\max}$ as the maximum voltage and set it to V when finding the max acceleration

Define $\alpha_{\max}$ as the maximum acceleration and set it to α when finding the max acceleration

$I_{\text{clamp}} = \min(\max(I, I_{\min}), I_{\max})$ by the definition of clamping

$I_{\text{clamp}} = \min(\max(\left(\frac{E_{mf} - V_{\max}}{-R}\right), I_{\min}), I_{\max})$ by substitution

$I_{\text{clamp}} = \min(\max(\left(\frac{\omega \kappa_e - V_{\max}}{-R}\right), I_{\min}), I_{\max})$ by substitution

Define $V_{\max} = V$ as the max voltage

$I_{\text{clamp}} = \min(\max(\left(\frac{\omega \kappa_e - V_{\max}}{-R}\right), I_{\min}), I_{\max})$ by substitution

Restrict I by $I_{\text{clamp}} = I$ by the note

$-J\alpha_{\max} = B\omega + \tau_{load} \text{sign}(\omega) - \tau$ by Subtraction

$\alpha_{\max} = \frac{B\omega + \tau_{load} \text{sign}(\omega) - \tau}{-J}$ by Division

$\alpha_{\max} = \frac{B\omega + \tau_{load} \text{sign}(\omega) - I \kappa_t}{-J}$ by Substitution

$\alpha_{\max} = \frac{B\omega + \tau_{load} \text{sign}(\omega) - I_{\text{clamp}} \kappa_t}{-J}$ by Substitution

$\alpha_{\max} = \frac{I_{\text{clamp}} \kappa_t - B\omega - \tau_{load} \text{sign}(\omega)}{J}$ by Substitution

Define a function α by $\alpha_{\max} = \frac{I_{\text{clamp}} \kappa_t - B\omega - \tau_{load} \text{sign}(\omega)}{J}$

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Find $\omega(t)$ given that the motor is at a Constant Voltage

$$\alpha_{\max}(\omega) = \frac{V_{\max} - K_v \omega - \text{sgn}(\omega) K_s}{K_a}$$

Let $s = \text{sgn}(\omega) \in \{1, -1\}$

$$\frac{d\omega}{dt} = \frac{V_{\max} - K_v \omega - s K_s}{K_a}$$

$$\frac{d\omega}{dt} + \frac{K_v}{K_a} \omega = \frac{V_{\max} - s K_s}{K_a}$$

$$I(t) = e^{\int \frac{K_v}{K_a} dt}$$

$$I(t) = e^{\frac{K_v}{K_a} t}$$

$$\omega(t) = \frac{1}{I(t)} \left(\int_0^t I(\tau) Q d\tau + I(0) \omega_0 \right)$$

$$\omega(t) = \frac{1}{e^{\frac{K_v}{K_a} t}} \left(\int_0^t e^{\frac{K_v}{K_a} \tau} \frac{V_{\max} - s K_s}{K_a} d\tau + \omega_0 \right)$$

$$\omega(t) = \frac{1}{e^{\frac{K_v}{K_a} t}} \left(\frac{V_{\max} - s K_s}{K_a} \frac{K_a}{K_v} \left(e^{\frac{K_v}{K_a} t} - 1 \right) + \omega_0 \right)$$

$$\omega(t) = \frac{V_{\max} - s K_s}{K_v} \frac{e^{\frac{K_v}{K_a} t} - 1}{e^{\frac{K_v}{K_a} t}} + \omega_0 \left(e^{-\frac{K_v}{K_a} t} \right)$$

$$\omega(t) = \frac{V_{\max} - s K_s}{K_v} + \left(\omega_0 - \frac{V_{\max} - s K_s}{K_v} \right) e^{-\frac{K_v}{K_a} t}$$

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Define the plant (solve for the derivative of the state x where V is the desired input for the plant)

$$x := \omega$$

$$V := u$$

$$\dot{x} = \dot{\omega} \text{ by the derivative}$$

$$\dot{x} = \alpha \text{ by substitution}$$

$$\therefore \text{Find } \dot{x}$$

$$V_{\text{ff}} = K_a \alpha_{\text{ref}} + K_v \omega_{\text{ref}} + K_s \text{sign}(\omega_{\text{ref}}) \text{ because it is given}$$

$$V_{\text{ff}} := u \text{ as the voltage is the input}$$

$$\alpha_{\text{ref}} := \alpha$$

$$\omega_{\text{ref}} := \omega$$

$$u = K_a \dot{x} + K_v x + K_s \text{sign}(x) \text{ by substitution}$$

$$K_a \dot{x} = u - K_v x - K_s \text{sign}(x) \text{ by subtraction}$$

$$\dot{x} = \frac{u - K_v x - K_s \text{sign}(x)}{K_a} \text{ by division}$$

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Find a continuous state-space model in the form

$$\dot{x} = Ax + Bu$$

$$y = Cx + Du$$

$$V_{\text{ff}} = K_a \alpha_{\text{ref}} + K_v \omega_{\text{ref}} + K_s \text{sign}(\omega_{\text{ref}}) \text{ because it is given}$$

$$u := V_{\text{fb}}$$

$$V := V_{\text{ff}} + V_{\text{fb}}$$

$$V = V_{\text{ff}} + u$$

$$\dot{x} = \frac{V - K_v x - K_s \text{sign}(x)}{K_a} \text{ because it is given and } V \text{ is the desired input for the plant}$$

$$e := x - x_{\text{ref}} \text{ for } x \text{ in the feedforward}$$

$$\dot{e} = \dot{x} - \dot{x}_{\text{ref}}$$

$$\dot{e} + \dot{x} = \frac{V - K_v x - K_s \text{sign}(x)}{K_a} + \dot{e} \text{ by addition}$$

$$\dot{x} - \dot{x}_{\text{ref}} + \dot{x} = \frac{V_{\text{ff}} + u - K_v \omega_{\text{ref}} - K_s \text{sign}(\omega_{\text{ref}})}{K_a} + \dot{x} - \dot{x}_{\text{ref}} \text{ by substitution}$$

$$\dot{x} - \dot{x}_{\text{ref}} + \dot{x} = \frac{K_a \alpha_{\text{ref}} + K_v \omega_{\text{ref}} + K_s \text{sign}(\omega_{\text{ref}}) + u - K_v \omega_{\text{ref}} - K_s \text{sign}(\omega_{\text{ref}})}{K_a} + \dot{x} - \dot{x}_{\text{ref}} \text{ by substitution}$$

$$\dot{x} - \dot{x}_{\text{ref}} = \frac{K_a \alpha_{\text{ref}} + K_v \omega_{\text{ref}} + K_s \text{sign}(\omega_{\text{ref}}) + u - K_v \omega_{\text{ref}} - K_s \text{sign}(\omega_{\text{ref}})}{K_a} - \dot{x}_{\text{ref}} \text{ by subtraction}$$

$$\dot{e} = \frac{u - K_v (\omega - \omega_{\text{ref}}) - K_s (\text{sign}(\omega) - \text{sign}(\omega_{\text{ref}})) + K_a \alpha_{\text{ref}}}{K_a} - \dot{\omega}_{\text{ref}} \text{ by substitution}$$

$$\dot{e} = \frac{u - K_v (x - x_{\text{ref}}) - K_s (\text{sign}(x) - \text{sign}(x_{\text{ref}})) + K_a \alpha_{\text{ref}}}{K_a} - \alpha_{\text{ref}} \text{ by substitution}$$

$$\text{Assume } \text{sign}(x) = \text{sign}(x_{\text{ref}}) \therefore \text{sign}(x) - \text{sign}(x_{\text{ref}}) = 0$$

$$\dot{e} = \frac{u - K_v e}{K_a} \text{ by substitution}$$

$$\dot{e} = -\frac{K_v}{K_a} e + \frac{1}{K_a} u \text{ by substitution}$$

$$x := e \quad A := -\frac{K_v}{K_a} \quad B := \frac{1}{K_a}$$

$$y := e \Rightarrow C = 1 \text{ and } D = 0$$

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Given

$$A_d = e^{A_c T}$$

$$B_d = \int_0^T e^{A_c \tau} d\tau B_c$$

$$C_d = C_c$$

$$D_d = D_c$$

$$A_c = -\frac{K_v}{K_a}$$

$$B_c = \frac{1}{K_a}$$

$$C_c = 1$$

$$D_c = 0$$

Find a discrete state-space model in the form

$$x_{k+1} = Ax_k + Bu_k$$

$$y_k = Cx_k + Du_k$$

$$A_d = e^{-\frac{K_v}{K_a} T} \text{ by substitution}$$

$$B_d = \frac{1}{K_a} \int_0^T e^{-\frac{K_v}{K_a} \tau} d\tau$$

$$B_d = -\frac{1}{K_a} \frac{K_a}{K_v} e^{-\frac{K_v}{K_a} \tau} \Big|_0^T$$

$$B_d = \frac{1}{K_v} \left(1 - e^{-\frac{K_v}{K_a} T} \right)$$

$$C_d = 1 \text{ by substitution}$$

$$D_d = 0 \text{ by substitution}$$

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Given

$$u_k^* = \arg \min_{u_k} \sum_{k=0}^{\infty} (x_k^T Q x_k + u^T R u_k)$$

subject to $x_{k+1} = Ax_k + Bu_k$ Find K based on Q and R in LQR

$$A^T S A - S - A^T S B (R + B^T S B)^{-1} B^T S A + Q = 0 \text{ by the discrete algebraic Riccati equation}$$

$$K = (R + B^T S B)^{-1} B^T S A \text{ by the discrete LQR result solved by the discrete algebraic Riccati equation}$$

$$A := e^{-\frac{K_v}{K_a} T} := a$$

$$B := \frac{1}{K_v} \left(1 - e^{-\frac{K_v}{K_a} T} \right) := b$$

$$C = 1 \text{ by substitution}$$

$$D = 0 \text{ by substitution}$$

$$a^2 S - S - \frac{a^2 b^2 S^2}{R + b^2 S} + Q = 0 \text{ by substitution}$$

$$a^2 S R + a^2 b^2 S^2 - S R - b^2 S^2 + Q R + Q b^2 S - a^2 b^2 S^2 = 0 \text{ by multiplication}$$

$$a^2SR - SR - b^2S^2 + QR + Qb^2S = 0 \text{ by substitution}$$

$$b^2S^2 + (R(1 - a^2) - Qb^2)S - QR = 0 \text{ by substitution}$$

$Q, R, b^2 > 0 \therefore 4b^2QR > 0 \therefore \sqrt{(R(1 - a^2) - Qb^2)^2 + 4b^2QR} > |R(1 - a^2) - Qb^2| \therefore$ the smaller root is negative which is not possible as $S > 0$ so only the positive root is valid

$$S = \frac{Qb^2 - R(1 - a^2) + \sqrt{(R(1 - a^2) - Qb^2)^2 + 4b^2QR}}{2b^2} \text{ by the quadratic formula}$$

$$K = \frac{abS}{R + b^2S} \text{ by substitution}$$